

The Noise on Ultraviolet Astronomical Images Captured by Back-illuminated Silicon Sensors

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ABSTRACT

Ultraviolet (UV) astronomy currently relies on back-illuminated silicon imaging sensors. While the noise in these sensors is typically assumed to be dominated by photon shot noise, measurements from HST’s Wide Field Camera 3 (WFC3) and the Interface Region Imaging Spectrograph (IRIS) reveal a significant discrepancy: the noise measured in the ultraviolet (UV) is systematically lower than theoretical predictions. We propose that this discrepancy is caused by partial charge collection (PCC), whereby a fraction of photogenerated electron-hole pairs recombine before they can be measured. We present a simple theoretical model, valid for wavelengths from 1 Å to 10 000 Å, that incorporates the effects of PCC and shows better agreement with the noise measurements from both WFC3 and IRIS, resolving the previously unexplained discrepancy. This finding implies that the signal-to-noise ratio achievable with these sensors is higher in the UV than previously understood, potentially impacting both future instrument proposals and the uncertainties of our current UV imagery. At about 2000 Å to 3500 Å, PCC is the dominant noise source in the high-signal limit.

Keywords: Astronomical instrumentation (799); Astronomical detectors (84); Ultraviolet astronomy (1736)

1. INTRODUCTION

Back-illuminated, silicon-based image sensors such as charge-coupled devices (CCDs) and complementary metal-oxide-semiconductor (CMOS) sensors are ubiquitous in ultraviolet (UV) astronomy, and are currently used in many UV instruments such as the Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012), the Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014), and HST’s Wide Field Camera 3 (WFC3) (R. A. Kimble et al. 2008). Predicting the noise measured by these sensors is important both for quantifying the uncertainty on current astronomical observations as well as for evaluating the performance of proposed future instruments.

In the high-signal limit (that is, neglecting readout noise), the largest source of noise measured by these sensors is thought to be shot noise (R. A. Stern et al. 1986). This noise is due to the quantized nature of light (W. Schottky 1918) and is not a property of the sensor *per se*, but its magnitude depends on the number of measured photons, which is determined by the sensor’s geometry and optical constants. Another well-known source of noise inherent to silicon sensors is Fano noise

(U. Fano 1947), the random variation of the number of electrons produced per interacting photon. This noise source is small (D. Rodrigues et al. 2023), and often ignored entirely, but it can be combined with the shot noise to form the traditional estimate of the total noise (R. A. Stern et al. 1986; J. R. Janesick 2001).

This traditional model of shot noise and Fano noise is very accurate in the visible (M. Marinelli & J. Green 2024) and X-ray (S. Ishikawa et al. 2011; P. S. Athiray et al. 2020; B. D. Schwab et al. 2020) wavelength regimes. However, this model appears to be less accurate in the UV. For example, J. P. Wülser et al. (2018) showed that IRIS measured *less* noise in the far ultraviolet (FUV) channel than the traditional model would suggest and similar results were found in the near ultraviolet (NUV) by WFC3 (T. Borders et al. 2010). This discrepancy indicates that the traditional model is incomplete, and that a more detailed model is needed to predict the noise measured by these sensors in the UV.

In this work, we aim to resolve the UV noise discrepancy by developing a noise model which adds two additional effects to the traditional model. The first effect is partial charge collection (PCC), which occurs when electron-hole pairs recombine before they can be measured by the sensor (R. A. Stern et al. 1994; J. R. Janesick 2001). Noise due to PCC is a well-known effect, for example D. Rodrigues et al. (2023) accounted

for PCC in their measurement of the Fano noise in soft X-ray (SXR), but its influence on the noise measured in UV has been rarely modeled in detail. The second effect is charge diffusion (or charge spreading), the tendency for photoelectrons to migrate into neighboring pixels (J. R. Janesick 2001), a mechanism suggested by both J. P. Wülser et al. (2018) and T. Borders et al. (2010) to explain the discrepancies they measured. We will model the contribution of each of these effects to the total noise and compare our predicted noise to the noise measured by IRIS and WFC3. We will also provide noise predictions for AIA and the Multi-slit Solar Explorer (MUSE) (B. De Pontieu et al. 2020) which can be used for estimating the uncertainty of observations captured by these instruments. Finally, we have made available a reference implementation of our noise model in Python which is available to be installed from the Python Package Index (PyPI).

2. SENSOR MODEL

In this work, we will use the back-illuminated CCD model described in R. A. Stern et al. (1994) as a basis for our sensor model to make it directly comparable with previous works in the literature. Figure 1 is a schematic of our sensor model and it shows the light-sensitive region of the sensor to be an epitaxial silicon layer with a thickness D . A fraction of the incident photons are either absorbed in the oxide layer (thickness δ) or reflected at the first or second interface. The illuminated side of the epitaxial layer is considered to have a PCC region of thickness W , where electron-hole pairs can recombine before being measured, and there is a depletion region of thickness z_d on the non-illuminated side.

In Section 2.1 we will calculate the signal predicted by this sensor model using the method described in R. A. Stern et al. (1994), adapted to more recent measurements. In Section 2.2 we will calculate the noise predicted by this sensor model in a self-consistent manner, and in Section 2.3 we will model the charge diffusion of this sensor which is needed for a complete description of the noise. Throughout this section we will adopt the convention that unprimed variables represent quantities incident on the sensor, primed variables represent quantities internal to the sensor, and the double-primed variables represent quantities measured by the sensor.

2.1. Signal

The average number of photoelectrons measured per incident photon is known as the quantum efficiency (QE). This is a common performance metric for measuring sensor sensitivity and it is given in J. R. Janesick

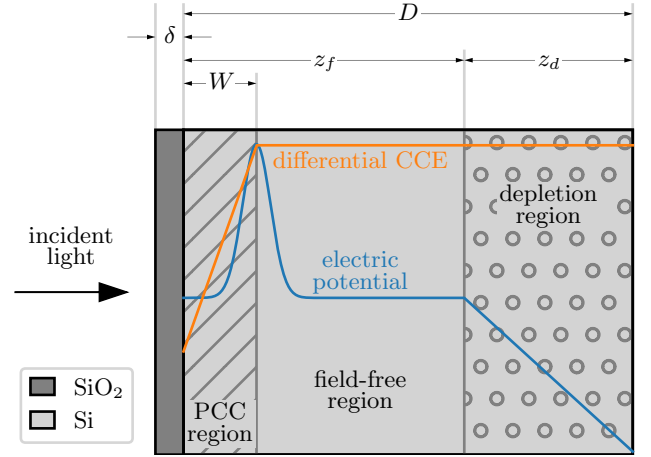


Figure 1. A schematic (not to scale) of the back-illuminated sensor model used in this work, labeled with the thicknesses of each layer. Overplotted is a *qualitative* example of the electric potential within the sensor that motivates the differential CCE.

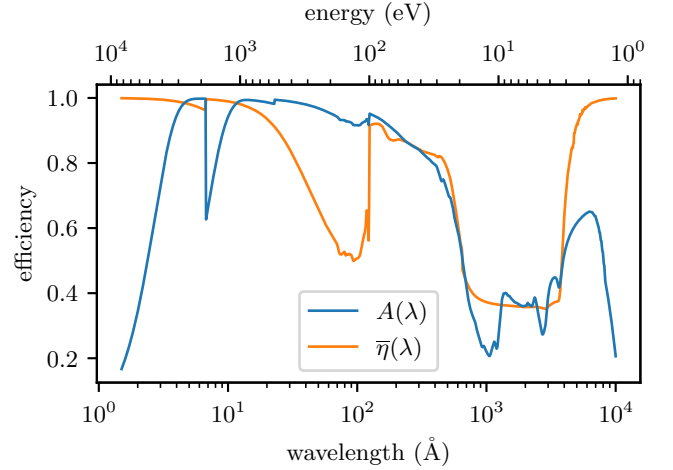


Figure 2. The fraction of incident light absorbed by the light-sensitive silicon layer and the CCE as a function of wavelength for the parameters inferred from the J. Heymes et al. (2020) measurements.

(2001) as

$$\text{QE}(\lambda, T) = \left\langle \frac{N_e''}{N_\gamma} \right\rangle = A(\lambda) \bar{\eta}(\lambda, T) \bar{\eta}(\lambda), \quad (1)$$

where λ is the wavelength of the incident photons in vacuum, T is the temperature of the sensor, $\langle X \rangle = \bar{X}$ denotes the expected value of a random variable X , N_e'' is the number of electrons measured by the sensor, N_γ is the total number of photons incident on the sensor, $A(\lambda)$ is the absorbance of the light-sensitive layer, $\bar{\eta}(\lambda, T)$ is the average quantum yield, the number of photoelectrons generated per absorbed photon, and $\bar{\eta}(\lambda)$ is the

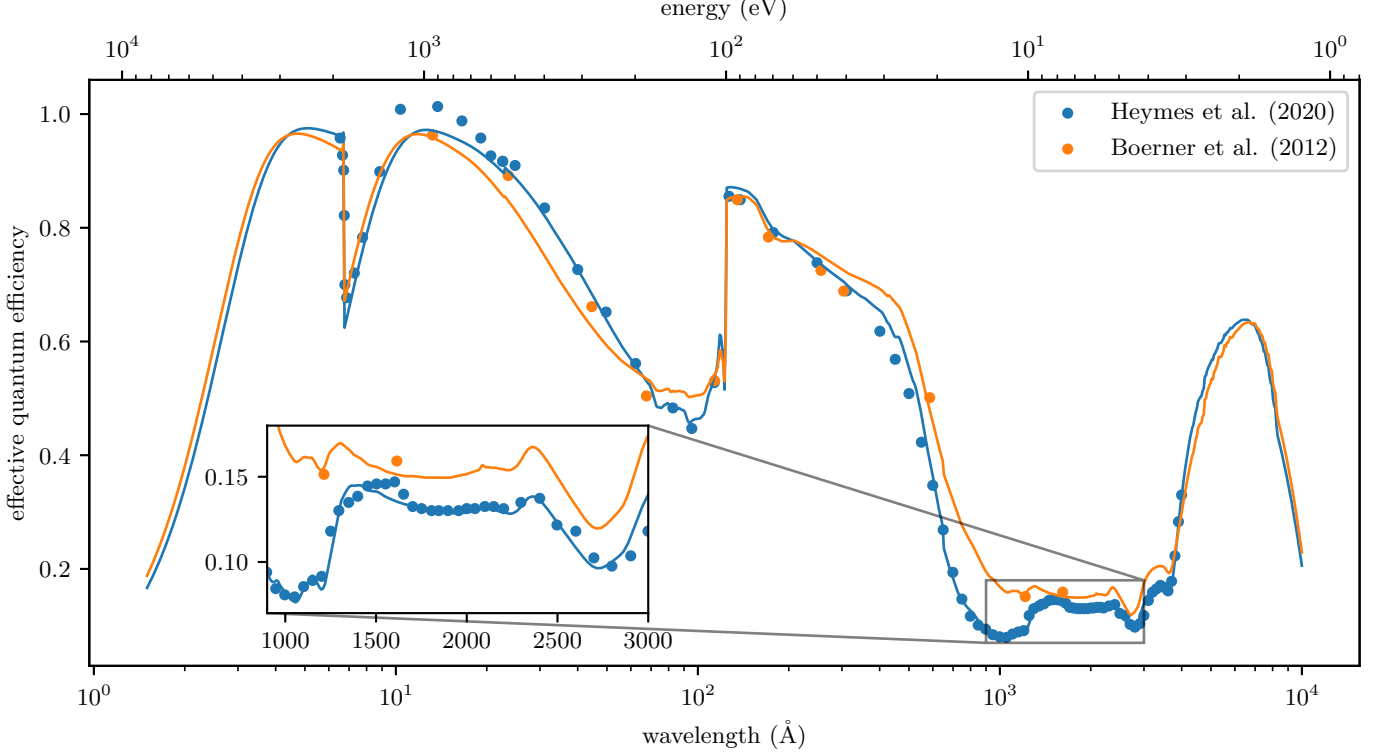


Figure 3. A comparison of $\text{EQE}(\lambda)$ measured by [P. Boerner et al. \(2012\)](#) and [J. Heymes et al. \(2020\)](#), along with the best-fit models described in Table 1 plotted as lines with the same color as the data. The inset zooms into the ultraviolet to better visualize the difference between the measurement and model in this range.

Table 1. The sensor model parameters which best fit the measurements in [P. Boerner et al. \(2012\)](#) and [J. Heymes et al. \(2020\)](#).

Source	implementation	η_0	W (nm)	δ (nm)	D (μm)	substrate roughness (nm)
P. Boerner et al. (2012)	<code>e2v_ccd203</code>	0.452	335	1.4	16	1.1
J. Heymes et al. (2020)	<code>e2v_ccd97</code>	0.334	153	4.6	14	4.8

average charge-collection efficiency (CCE), the fraction of generated photoelectrons measured by the sensor.

The fraction of incident energy that is absorbed by the light sensitive layer is the absorbance, $A(\lambda)$, which is reduced by reflections at each interface, absorption in the oxide layer (UV), or by penetration through the device (SXR and infrared (IR)). $A(\lambda)$ can be determined from the optical constants of Si and SiO_2 using, for example, the popular IMD code ([D. L. Windt 1998](#)). For this work, we used our Python library, `optika` ([R. T. Smart & C. C. Kankelborg 2024](#)), which uses the transfer matrix method described in [P. Yeh \(1988\)](#) with the optical constants from [E. D. Palik \(1997\)](#), [B. Henke et al. \(1993\)](#), and [L. V. R. de Marcos et al. \(2016\)](#) to compute the electric field for every interface in the sensor. In Figure 2 we have plotted $A(\lambda)$ for the [J. Heymes et al. \(2020\)](#) parameters in Table 1.

In [R. A. Stern et al. \(1994\)](#), the authors assume no reflections from the unilluminated side of the sensor for simplicity. In this work, we compute the total change in Poynting flux into and out of the light-sensitive region of the sensor to determine $A(\lambda)$. This treatment introduces interference effects for infrared wavelengths, which can be seen on the right side of Figure 2.

Determining $\bar{n}(\lambda, T)$ over the entire wavelength range considered in this study is an area of ongoing research ([G. W. Fraser et al. 1994](#); [J. Geist 1996](#); [F. Scholze et al. 1998](#); [J. Fang et al. 2019](#), just to name a few). This work will use the quantum yield model developed by [K. Ramanathan & N. Kurinsky \(2020\)](#) which uses a phenomenological model of impact ionization to bridge the “UV gap”, an area where direct measurements of the quantum yield are not yet available. They give the av-

average quantum yield as

$$\bar{n}(E, T) = \begin{cases} \sum_{n=0}^{\infty} n p_n(E, T), & 0 < E \leq 50 \text{ eV} \\ E/\epsilon_{eh}(T), & 50 \text{ eV} < E < \infty, \end{cases} \quad (2)$$

where E is the energy of the incident photon, n is the actual quantum yield, $p_n(E, T)$ is a table of pair-creation probabilities which is provided in their supplementary material,

$$\epsilon_{eh}(T) = 1.7E_g(T) + (0.084 \text{ eV}^{-1})A + 1.3 \text{ eV}, \quad (3)$$

is the asymptotic mean energy per electron-hole pair, $A = 5.2 \text{ eV}^2$, and

$$E_g(T) = 1.1692 \text{ eV} - \frac{(4.9 \times 10^{-4} \text{ eV K}^{-1}) T^2}{T + 655 \text{ K}}, \quad (4)$$

is the bandgap energy of silicon.

In [R. A. Stern et al. \(1994\)](#), the average CCE is expressed in terms of a differential CCE, $\eta(z)$, which is the fraction of photoelectrons collected for a photon absorbed at a depth z into the epitaxial layer. The average CCE is then the first moment of the differential CCE weighted by the probability of absorbing a photon at a depth z ,

$$\bar{\eta}(\lambda) = \frac{\int_0^{\infty} \eta(z) e^{-\alpha(\lambda)z} dz}{\int_0^{\infty} e^{-\alpha(\lambda)z} dz}, \quad (5)$$

where $\alpha(\lambda)$ is the absorption coefficient of silicon for the given wavelength.

In principle, $\eta(z)$ is a function of the exact implant profile which is usually impractical to measure, but see [R. A. Stern et al. \(2004\)](#); [P. Boerner et al. \(2012\)](#) for a case where the authors did have a measurement of the implant profile provided by the manufacturer. In [R. A. Stern et al. \(1994\)](#), the authors instead adopt a piecewise-linear approximation of the differential CCE,

$$\eta(z) = \begin{cases} \eta_0 + (1 - \eta_0)z/W, & 0 < z < W \\ 1, & W < z < \infty \end{cases} \quad (6)$$

where η_0 is the differential CCE at the back surface of the sensor. Plugging Equation 6 into Equation 5 yields an expression for the CCE,

$$\bar{\eta}(\lambda) = \eta_0 + \left(\frac{1 - \eta_0}{\alpha W} \right) (1 - e^{-\alpha W}), \quad (7)$$

which can be used in Equation 1 to determine the QE. In Figure 2 we have plotted the average CCE for the [J. Heymes et al. \(2020\)](#) parameters in Table 1.

In [R. A. Stern et al. \(1994\)](#), the authors define an effective QE as

$$\text{EQE}(\lambda) = A(\lambda) \bar{\eta}(\lambda), \quad (8)$$

which is found by taking the ratio of the current measured by the sensor to the current measured by a calibrated photodiode and is the quantity that is typically measured when calibrating an image sensor ([R. A. Stern et al. 1994, 2004](#); [P. Boerner et al. 2012](#)). In Figure 3, we have plotted the measured, effective QE for two sources: [P. Boerner et al. \(2012\)](#) which measured the AIA CCDs at a few discrete wavelengths, and [J. Heymes et al. \(2020\)](#) which measured a Teledyne e2v CCD97 sensor over a wide wavelength range with high resolution using a monochromator.

Using the Nelder-Mead minimization algorithm ([F. Gao & L. Han 2010](#)), we found the free parameters of our model which best fit the data in [P. Boerner et al. \(2012\)](#) and [J. Heymes et al. \(2020\)](#). These models are plotted as solid lines in Figure 3, and the corresponding values of the free parameters are shown in Table 1. Throughout the remainder of this work we will use the model which best fits the [J. Heymes et al. \(2020\)](#) data.

2.2. Noise

Our noise model will consider three sources: shot noise from the quantized nature of the photons striking the sensor, Fano noise due to inherent randomness in the process which converts photons to electrons, and noise due to electrons randomly recombining before they can be measured by the sensor. This section will describe the statistics of each noise source and demonstrate a procedure which can simulate the noise measured by our model sensor for a given number of incident photons.

Throughout this work, we will measure noise in terms of a variance-to-mean ratio (VMR),

$$F(X) = \frac{\text{Var}(X)}{\bar{X}}, \quad (9)$$

where $\text{Var}(X)$ is the variance of X . Using the VMR instead of the signal-to-noise ratio (SNR) to express the noise is convenient since it is constant as a function of signal for the distributions studied here. For example, the VMR of a Poisson random variable is always unity since its variance and expectation value are equal. The VMR is not dimensionless (it has the same units as X), so we must take care to interpret the VMR in terms of the correct units. For a constant multiple of a random variable, aX ,

$$F(aX) = aF(X). \quad (10)$$

We will use this relationship throughout this work to convert from incident photons to measured electrons and vice versa.

In Figure 4 we have plotted the VMR for the noise sources considered in this study in two different units: number of incident photons and number of measured

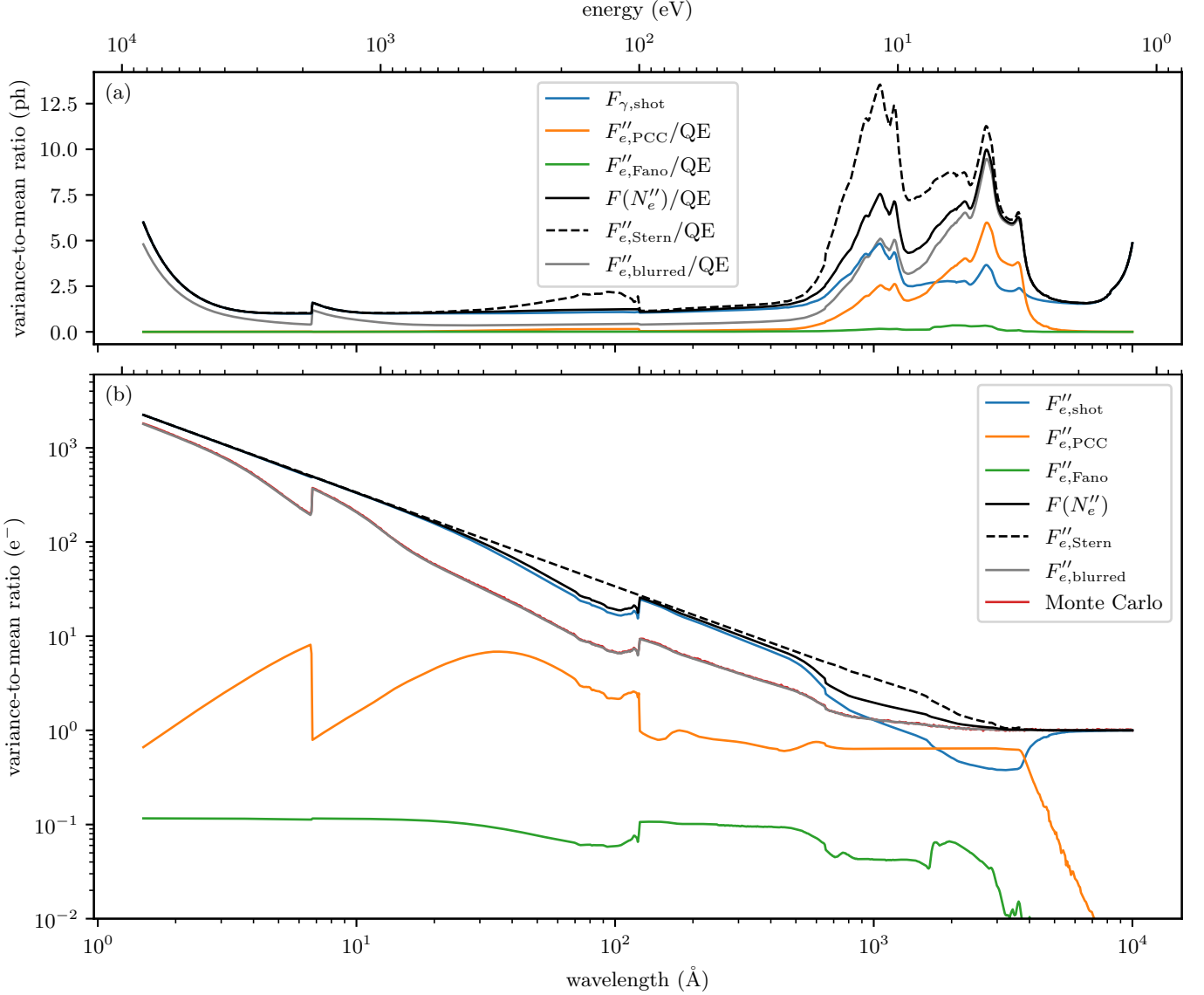


Figure 4. The total and component-wise VMR of our sensor model expressed in two different units: photons incident on the sensor (top) and electrons measured by the sensor (bottom). Plotted for comparison is the VMR of the [R. A. Stern et al. \(1986\)](#) noise model (dashed) and a Monte Carlo simulation of $F(N''_e)$ in red.

electrons. Figure 4a is useful when *designing* an instrument since the number of photons incident on the sensor can be determined by the radiance of the source and the effective area of the instrument. Figure 4b is useful when *calibrating* an instrument since we directly measure these electrons. The details of the calculations for Figure 4 are presented in the subsections below.

2.2.1. Shot Noise

Photon shot noise is often assumed to be the leading noise contributor in UV astronomy ([R. A. Stern et al. 1986](#); [J. R. Lemen et al. 2012](#); [B. De Pontieu et al. 2014](#)). If the expected number of photons that interact with the

light-sensitive layer is

$$\bar{N}'_\gamma = A\bar{N}_\gamma. \quad (11)$$

then the actual number of interacting photons, N'_γ , is sampled from the Poisson distribution,

$$N'_\gamma \leftarrow \text{Poisson}(\bar{N}'_\gamma). \quad (12)$$

As mentioned in the previous section, the VMR of this process is

$$F(N'_\gamma) = 1, \quad (13)$$

but this in terms of the number of absorbed photons, which is internal to the sensor. In terms of the expected number of incident photons given the number

of absorbed photons the VMR is

$$F_{\gamma, \text{shot}} \equiv F\left(\frac{N'_\gamma}{A}\right) = \frac{1}{A} \quad (14)$$

since A is the conversion factor between incident photons and absorbed photons. $F_{\gamma, \text{shot}}$ is plotted in Figure 4a in blue and demonstrates that this expression is nearly unity across almost the entire wavelength range of the sensor except in the UV, where the absorbance is poor (Figure 2). We can also express the VMR of the shot noise in terms of the number of measured electrons by using the QE as the conversion factor between the unprimed and double-primed variables,

$$F''_{e, \text{shot}} = \bar{n} \bar{\eta}. \quad (15)$$

This equation is plotted in Figure 4b in blue, where we can see that it increases linearly with decreasing wavelength since the average quantum yield is increasing.

2.2.2. Fano Noise

The energy resolution of silicon detectors is ultimately limited due to Fano noise (U. Fano 1947), the random variation of the quantum yield n . Fano noise is usually expressed in terms of the Fano factor, $\mathcal{F} = F(n)$. Silicon is commonly accepted to have $\mathcal{F} \approx 0.1$ (J. R. Janesick 2001). In part due to variations of the Fano noise as a function of wavelength and temperature (G. W. Fraser et al. 1994), there is some disagreement in the literature around a more precise value for $\mathcal{F}$ (G. W. Fraser et al. 1994; B. Lowe 1997; M. N. Mazziotta 2008; I. V. Kotov et al. 2018; D. Rodrigues et al. 2021, 2023, & references therein). However, because the Fano noise is so small compared to the other noise sources considered in this study, the precise form of this noise model does not appreciably influence our conclusions.

As we mentioned in Section 2.1, this work uses the quantum yield model of K. Ramanathan & N. Kurinsky (2020). They provide the probability mass function (PMF) of the quantum yield distribution, $p_n(E, T)$, which we can use directly to generate random samples of the quantum yield for $E \leq 50$ eV. For $E > 50$ eV, we will assume that the quantum yield is a Gaussian distribution centered around $\bar{n}(\lambda, T)$, with standard deviation $\sqrt{\bar{n}(\lambda, T) \mathcal{F}(T)}$, where $\mathcal{F}(T)$ is the asymptotic Fano factor given by K. Ramanathan & N. Kurinsky (2020) as

$$\mathcal{F}(T) = (-0.028 \text{ eV}^{-1}) E_g(T) + (0.0015 \text{ eV}^{-2}) A + 0.14. \quad (16)$$

At 300 K, this corresponds to a Fano factor of approximately 0.114, which is generally consistent with the best-available measurement of the Fano factor by D. Rodrigues et al. (2021) at 6 keV.

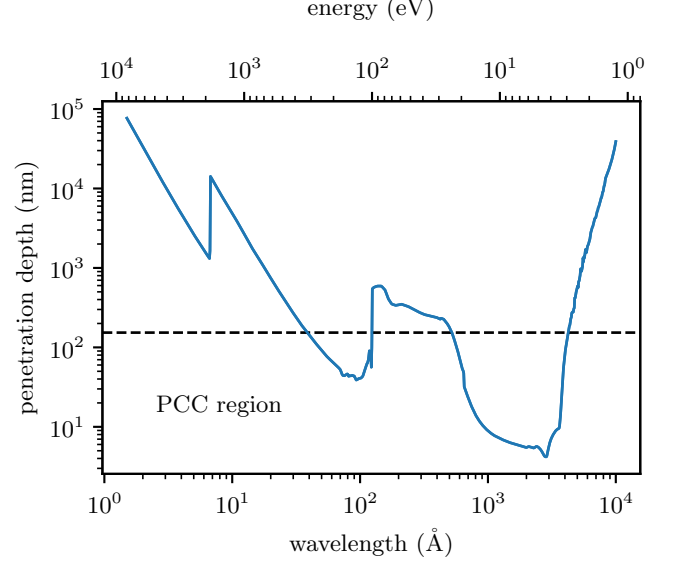


Figure 5. The penetration depth in silicon as a function of wavelength plotted against the depth of the PCC region.

$\mathcal{F}$ is a quantity internal to the sensor. The measured Fano factor is instead

$$F''_{e, \text{Fano}} = \bar{\eta} \mathcal{F} \quad (17)$$

since the conversion factor between the generated electrons and the measured electrons is the CCE. In Figure 4b we have plotted $F''_{e, \text{Fano}}$ in green. This shows that the Fano noise is the smallest contributor to the total noise across the entire wavelength range and is often at least an order of magnitude smaller than the other noise sources.

2.2.3. Partial Charge Collection Noise

In Figure 5, we have plotted the penetration depth in silicon vs. the thickness of the PCC region. We can see that there are two regions in the UV where the penetration depth is less than the thickness of the PCC region. In these regions, there is a significant chance that some of the electron-hole pairs will randomly recombine before being measured by the sensor. The stochastic nature of this process leads us to consider this as a new noise source, PCC noise.

In the R. A. Stern et al. (1994) charge-collection model, each photoelectron generated in the PCC region has a probability $\eta(z)$ of *not* recombining and subsequently being measured by the sensor. If n'_i is the quantum yield of the i th photon generated using the K. Ramanathan & N. Kurinsky (2020) model discussed in the previous section, we can express the *measured* quantum yield as

$$n''_i \leftarrow \text{Binomial}(n'_i, \eta(z_i)), \quad (18)$$

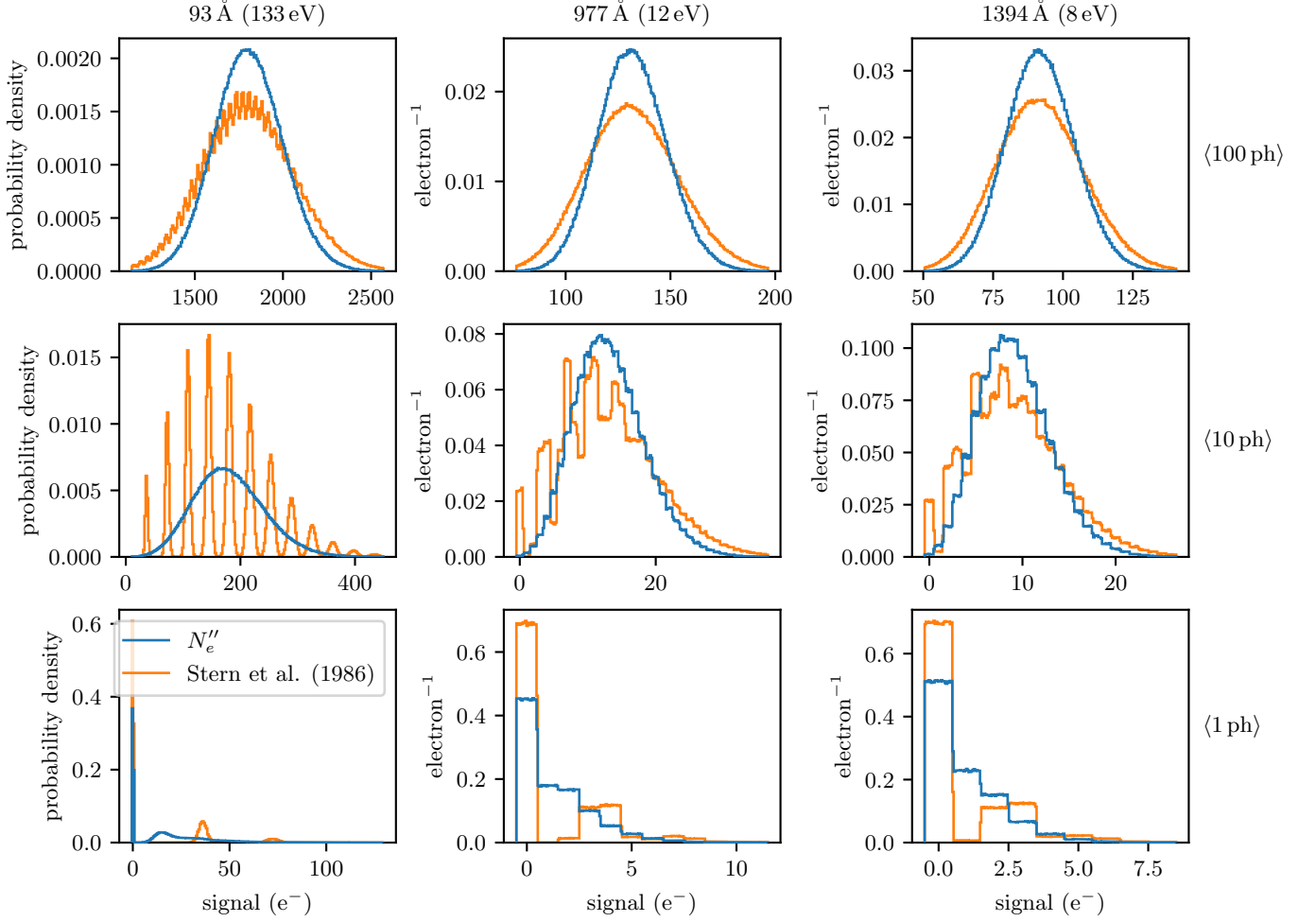


Figure 6. The probability distribution of the number of measured electrons for a given wavelength and expected number of absorbed photons calculated using 1000000 samples of Equation 19 and the R. A. Stern et al. (1986) noise model.

where z_i is the penetration depth of the i th photon and $\text{Binomial}(n, p)$ is a sample from the binomial distribution with n trials and probability of success p . We can use 18 to compute the total number of measured electrons given N'_γ absorbed photons as

$$N''_e = \sum_{i=0}^{N'_\gamma} n''_i. \quad (19)$$

This expression is intended to be used in an instrument forward model to sample the distribution of measured electrons given an expected number of incident photons.

The VMR of this process is non-trivial since Equation 19 is a sum of Binomial draws with completely independent n and p for each term. However, if we compute the VMR of the CCE by taking the second moment of Equation 6 in a similar fashion to Equation 5,

$$F(\eta) = \frac{2e^{-\alpha W}}{\bar{\eta}} \left(\frac{1 - \eta_0}{\alpha W} \right)^2 (\sinh(\alpha W) - \alpha W), \quad (20)$$

we can use the expressions given by heropup (2019) to find the VMR of only the Fano noise and PCC noise as

$$F''_{e,\text{sensor}} = 1 + (\mathcal{F} - 1)\bar{\eta} + (\bar{n} + \mathcal{F} - 1)F(\eta). \quad (21)$$

The contribution from only PCC noise is then

$$F''_{e,\text{PCC}} = F''_{e,\text{sensor}} - F''_{e,\text{Fano}}, \quad (22)$$

which is plotted in Figure 4 in orange. We can see that the PCC noise is usually very small compared to the shot noise, but in the UV from about 2000 Å to 3500 Å, it actually exceeds the shot noise and is the dominant contributor to the noise.

In Figure 6, we have plotted the PMFs of Equation 19 and the R. A. Stern et al. (1986) noise model to visualize the differences between these two distributions. Each model has been evaluated on a grid where each column of the grid represents a different wavelength (chosen to demonstrate the worst-case differences between the two models) and each row represents a different expected

number of absorbed photons. When the number of absorbed photons is low, like in the bottom two rows of Figure 6, we can see that the [R. A. Stern et al. \(1986\)](#) model (orange) has a comb-like appearance caused by the Fano noise slightly blurring the PMF of the photon shot noise. In contrast, our model shows much less of this comb pattern since the PCC noise tends to blur the distribution further into a single peak. As the number of photons increases, both distributions tend towards a Gaussian, but our model remains narrower.

To ease adoption of this model, we've provided a reference implementation of Equation 19 in Python, `optika.sensors.signal()`, which is designed to be simple to use for existing and future instrument pipelines.

2.3. Charge Diffusion

In most back-illuminated imaging sensors used for UV astronomy, the depletion region (the region with significant electric field) does not extend all the way from the gate structure through the device. As a result, there is a so-called field-free region near the back of the sensor where photoelectrons must undergo a random walk to find their way to the depletion region where they can then be conducted to the terminals and measured ([J. R. Janesick 2001](#)). This random walk generally leads to a loss of spatial resolution measured by the sensor since electrons can diffuse to adjacent pixels. It also leads to a reduction in the variance of a flat field measured by the sensor since the blurring due to this diffusion induces a correlation between neighboring pixels.

Using Monte Carlo modeling, [J. R. Janesick \(2001\)](#) found the following expression for the standard deviation of the charge diffusion kernel:

$$\sigma(z) = \begin{cases} z_f \sqrt{1 - z/z_f}, & 0 < z < z_f \\ 0, & z_f < z < D, \end{cases} \quad (23)$$

where z is the distance from the back surface at which the photon is absorbed, $z_f = D - z_d$ is the thickness of the field-free region of the sensor, and z_d is the thickness of the depletion region. Using Equation 23, we can find the average variance of the charge diffusion kernel by taking a mean across the entire thickness of the sensor weighted by the probability of a photon being absorbed at that depth,

$$\overline{\sigma^2} = \frac{z_f (\alpha z_f + e^{-\alpha z_f} - 1)}{\alpha (1 - e^{-\alpha D})}. \quad (24)$$

The thickness of the depletion region or the field-free region is difficult to measure, and depends on the voltage applied to the sensor and the charge collected at the terminals ([R. A. Stern et al. 2004](#)).

However, [R. A. Stern et al. \(2004\)](#) did estimate the size of the charge diffusion kernel, for two discrete wavelengths, of a 15 μm -thick (100 $\Omega\text{-cm}$ resistivity) CCD for the GOES Soft X-ray Imager. We can use these measurements to estimate the size of the depletion region and model the size of the charge diffusion kernel as a function of wavelength. [R. A. Stern et al. \(2004\)](#) did not directly measure the size of the charge diffusion kernel, instead they measured a quantity they named the mean charge capture (MCC), the fraction of charge captured by the central pixel. Naively, the MCC would be the integral of the charge diffusion kernel over the extent of a pixel. However, since a photon can strike anywhere within the central pixel, we need to convolve with a rectangle function the width of a pixel before integrating. If we assume that the charge diffusion kernel is a Gaussian with variance σ^2 , then the MCC is

$$m = \left[\frac{1}{\sqrt{\pi a}} (e^{-a} - 1) + \text{erf}(\sqrt{a}) \right]^2, \quad (25)$$

where $a = d^2/2\sigma^2$, d is the width of a pixel, and $\text{erf}(x)$ is the error function.

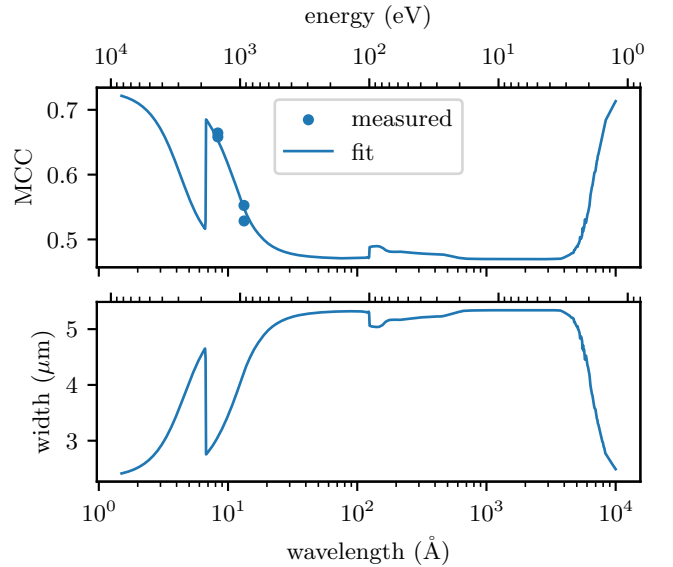


Figure 7. The top panel plots the MCC measured by [R. A. Stern et al. \(2004\)](#) and the fit of our model. The bottom panel shows the corresponding standard deviation of the charge diffusion kernel for the [J. Heymes et al. \(2020\)](#) sensor.

In the top panel of Figure 7, we have plotted a fit of Equation 25 to the measurements in [R. A. Stern et al. \(2004\)](#) which found $z_d = 8.7 \mu\text{m}$ best matched the data. In the lower panel of Figure 7, we have plotted the corresponding standard deviation of the charge diffusion kernel as a function of wavelength which predicts that

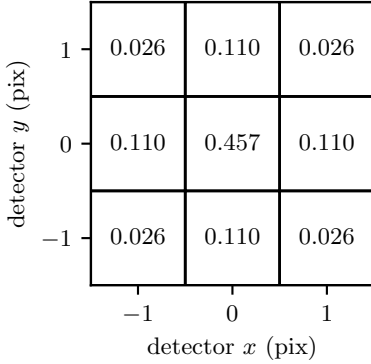


Figure 8. The charge diffusion kernel at 1400 Å convolved with a 13 μm IRIS pixel and integrated over the extent of each pixel.

the charge diffusion is reasonably constant over much of the SXR and ultraviolet wavelengths since the penetration depth is low in this regime.

To visualize the extent of this charge diffusion, we’ve included Figure 8, a 3 × 3 kernel where the value in each pixel has been computed in a manner similar to Equation 25, just with different limits of integration. We used the IRIS pixel size and a representative wavelength to demonstrate the worst-case scenario for IRIS. Convolution with this kernel with an image approximates the spatial extent of the charge diffusion, but it does not describe the process faithfully. A convolution assigns a fractional number of electrons to each pixel, whereas every electron is in fact collected by exactly one pixel, and it treats the displacement of each electron as independent, whereas the electrons liberated by a single photon share both an absorption depth and a sub-pixel origin. We therefore diffuse each electron individually, displacing it by $\mathcal{N}(0, \sigma^2(z_i))$ in each direction and rounding to the nearest pixel, which conserves charge exactly and preserves the correlation between electrons from the same photon. This is implemented by `optika.sensors.signal()`, and the kernel in Figure 8 was computed with `optika.sensors.kernel_diffusion()`, which can be used to visualize the charge diffusion for other wavelengths and pixel sizes.

Charge diffusion does not add a noise source of its own. It moves electrons between pixels without creating or destroying any, so it leaves the mean of a flat-field image unchanged, and it reduces the variance only by weakening the correlation between electrons that came from the same photon. Two electrons liberated by one photon share an absorption depth and a sub-pixel origin, so before diffusion they are always measured together; afterwards they are measured together only with prob-

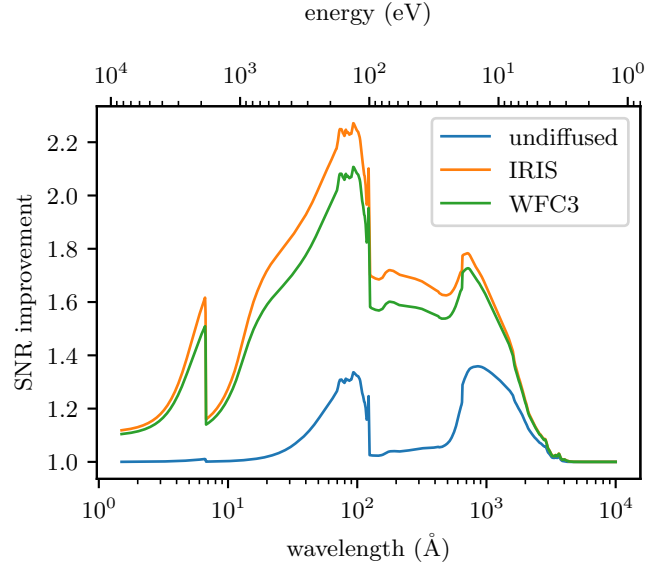


Figure 9. The SNR improvement predicted by our noise model compared to the [R. A. Stern et al. \(1986\)](#) model. In blue we have plotted the SNR improvement ignoring charge diffusion. In orange and green we have plotted the SNR improvement, including charge diffusion, for a flat-field image observed by IRIS and WFC3.

ability

$$\mathcal{P}(z) = p(\sigma(z)/d)^2, \quad (26)$$

$$p(s) = \text{erf}\left(\frac{1}{2s}\right) - \frac{2s}{\sqrt{\pi}} \left(1 - e^{-1/4s^2}\right),$$

where $p(s)$ is the probability that two electrons displaced independently from a common, uniformly distributed sub-pixel origin fall in the same column of pixels, and d is the width of a pixel. Weighting the photon-correlated term of Equation 21 by $\mathcal{P}(z)$ therefore gives the VMR in the presence of charge diffusion, which is the form implemented by `optika.sensors.vmr_signal()` and plotted for IRIS in Figure 4.

3. RESULTS AND DISCUSSION

Since $\text{EQE}(\lambda)$ can be interpreted as an efficiency, it is tempting to think that the expected number of measured photons and their variance is $\text{EQE} \times \bar{N}_\gamma$. However, using the effective QE in this way is equivalent to an all-or-nothing charge collection model where either all the electrons associated with an absorbed photon are measured or none of them are. This simple noise model is formalized in Equations 9 and 10 of [R. A. Stern et al. \(1986\)](#), which gives the VMR of the number of measured electrons as

$$F''_{e,\text{Stern}} = \bar{n} + \mathcal{F} \quad (27)$$

Table 2. The VMR predicted by our model (including charge diffusion) for prominent wavelengths in selected UV observatories, in both incident photon and measured electron units. Also included is the SNR improvement ratio between our model and the traditional noise model as well as the standard deviation of the charge diffusion kernel.

Instrument	wavelength ($\text{\AA}$)	VMR (ph)	VMR (e^-)	SNR improvement	diffusion width(μm)
AIA	94	0.43	6.96	2.27	5.32
	131	0.40	8.98	1.69	5.05
	171	0.42	6.72	1.72	5.15
	193	0.44	5.96	1.71	5.17
	211	0.45	5.53	1.70	5.17
	304	0.51	3.96	1.68	5.21
	335	0.53	3.64	1.67	5.21
	1600	3.83	1.15	1.42	5.34
	1700	4.51	1.12	1.35	5.34
IRIS	1330	3.22	1.20	1.52	5.34
	1400	3.23	1.19	1.49	5.34
	2796	9.21	1.02	1.08	5.34
	2832	8.93	1.02	1.09	5.34
MUSE	108	0.43	6.75	2.15	5.32
	171	0.42	6.72	1.72	5.15
	284	0.49	4.21	1.69	5.20
WFC3	2080	6.08	1.07	1.19	5.34
	2240	6.57	1.05	1.15	5.34
	2400	6.40	1.04	1.13	5.34
	2560	7.90	1.03	1.10	5.34
	2720	9.51	1.03	1.09	5.34
	2880	8.51	1.03	1.08	5.34
	3040	6.56	1.02	1.05	5.34
	4000	3.01	1.01	1.01	5.30

This is the noise model used by [T. Borders et al. \(2010\)](#) and [J. P. Wülser et al. \(2018\)](#). For comparison, the VMR predicted by our undiffused noise model is the sum of $F''_{e,\text{shot}}$ and $F''_{e,\text{sensor}}$,

$$F(N''_e) = 1 + (\bar{n} + \mathcal{F} - 1)(\bar{\eta} + F(\eta)). \quad (28)$$

In Figure 4, we have compared $F''_{e,\text{Stern}}$ (dashed) to our undiffused model, $F(N''_e)$ (black). Over most of the SXR and visible wavelengths the [R. A. Stern et al. \(1986\)](#) model is a good approximation of the undiffused model developed in this work since the penetration depth is deeper than the PCC region. However, in the UV, where the penetration depth is very shallow, [R. A. Stern et al. \(1986\)](#) overestimates the noise compared to our undiffused model.

It may seem paradoxical to add a noise source and then measure less total noise, but this is because more photons are measured in our model than the all-or-nothing model. Even though PCC introduces noise, it is

less noise than there would be if the photon was undetected. This is good news for designers of UV astronomical instruments since there is less noise than expected from the [R. A. Stern et al. \(1986\)](#) model in this wavelength range.

In Figure 9, we have plotted the ratio of the SNR predicted by our model to the SNR predicted by the traditional model. The blue line is the improvement ratio for the undiffused model and it shows that there are two regions, one from 30-100 $\text{\AA}$ and another from 500-2000 $\text{\AA}$, where the noise statistics deviate from the traditional model, in some cases by up to $\sim 30\%$. This figure can be used to quickly estimate the importance of PCC noise in a given wavelength range.

In Table 2, we have calculated the VMR in terms of incident photons and measured electrons for the target wavelengths of a few popular and upcoming solar instruments: AIA ([J. R. Lemen et al. 2012](#)), IRIS ([B. De](#)

Table 3. The ratio of the VMR of a UV flat-field image to the VMR of a visible-light flat-field image. The second column from the right is the value of this ratio measured by T. Borders et al. (2010) and J. P. Wülser et al. (2018). For IRIS this is the ratio of the inverse camera gains measured from photon-transfer curves, 12 photons per data number using a deuterium lamp at 1350 Å and 18 photons per data number using the onboard blue light-emitting diode (J. P. Wülser et al. 2018), which we take to be the 430 nm device flown on the GOES Soft X-ray Imager (R. A. Stern et al. 2004). The rightmost column is the value of this quantity predicted by our noise model (including charge diffusion).

Instrument	λ_{UV} (Å)	λ_{vis} (Å)	measurement	model
IRIS	1350	4300	1.50	1.19
WFC3	2080	4000	1.09	1.06
	2240	4000	1.09	1.05
	2400	4000	1.07	1.04
	2560	4000	1.04	1.03
	2720	4000	1.04	1.02
	2880	4000	1.03	1.02
	3040	4000	1.01	1.01

Pontieu et al. 2014), and MUSE (B. De Pontieu et al. 2020). The results show that for the extreme ultraviolet (EUV) channels, AIA and MUSE are nearly shot-noise-limited since the VMR in units of incident photons is near unity. Tabulated in the second-to-last column of Table 2 is the improvement factor between the VMR of the R. A. Stern et al. (1986) noise model and our noise model. These ratios show that the AIA 94 Å and 1600 Å, IRIS 1330 Å and 1400 Å, and MUSE 108 Å channels are predicted to have at least 20% more SNR than the traditional noise model would suggest. These results include the influence of charge diffusion, and we’ve also tabulated the size of the charge diffusion kernel in the last column, since it sets the scale over which the charge from a single photon is shared between neighboring pixels.

In Table 3, we’ve attempted to reproduce the measurements of J. P. Wülser et al. (2018) and T. Borders et al. (2010) by taking the ratio of the VMR of a simulated UV flat-field image to the VMR of a simulated visible flat-field image. The flat-field images were created by drawing samples from Equation 19 and then diffusing each measured electron individually, using the pixel size of the corresponding instrument. This table shows that the discrepancy discussed in the introduction is substantially reduced, though our model does not reproduce the measurements exactly. For WFC3, T. Borders et al. (2010) measured 1.09 at 2080 Å where the traditional

model expects about 1.7, and our model predicts 1.06, close to the measured value. For IRIS, J. P. Wülser et al. (2018) measured 1.5 at 1350 Å where the traditional model expects about 2, and our model predicts 1.19. This is much nearer the measurement than the traditional model, but it underestimates it, where the traditional model overestimates it. Our model therefore accounts for most of the reported discrepancy in both cases, but it overcorrects for IRIS. The remaining disagreement appears to be confined to our treatment of charge diffusion. Evaluating the same model without charge diffusion raises the predicted IRIS ratio to approximately the measured value, which suggests that the quantum yield and the PCC are not responsible.

Our diffusion model has three limitations, any of which could plausibly account for it. First, we have adopted the thickness of the depletion region fit to the measurements of R. A. Stern et al. (2004), which were made on a different sensor operated at a different voltage. The depletion thickness depends on both the resistivity of the silicon and the applied bias, so it need not carry over between sensors, and since the charge diffusion depends only on the difference between the substrate and depletion thicknesses, an error in one cannot be distinguished from an error in the other. A thicker depletion region implies a thinner field-free region and therefore less charge diffusion, which would raise our predicted ratios and improve the agreement for IRIS. Second, we model the charge cloud as a Gaussian, whereas G. G. Pavlov & J. A. Nousek (1999) show that the true radial distribution is peaked more strongly and has heavier tails, which would place more of the charge liberated by a single photon in a single pixel than we have assumed. Third, we neglect the further diffusion that the charge undergoes while drifting across the depletion region.

4. CONCLUSIONS AND FUTURE WORK

This work aims to resolve the discrepancy identified by T. Borders et al. (2010) and J. P. Wülser et al. (2018) in the predicted vs. observed noise measured by back-illuminated silicon imaging sensors. To accomplish this, we reconsidered the effects of PCC and charge diffusion on the noise by developing a simple model of a back-illuminated sensor based on the CCE introduced by R. A. Stern et al. (1994) and a charge diffusion model found by J. R. Janesick (2001). By analyzing these effects in a self-consistent manner, we were able to mostly resolve the discrepancy, and we propose that PCC is the most important effect to consider in the UV. The width of the charge diffusion kernel is nearly independent of wavelength, but its effect on the VMR is not,

since diffusion suppresses the photon-correlated term, which grows with the quantum yield.

Our sensor model predicts that there are two UV bands, 30-100 Å and 500-2000 Å, where PCC effects should be considered for a correct noise estimate. We evaluated our sensor model for some popular and upcoming UV astronomical instruments and found that there are a few channels in each instrument where our model implies that the SNR is better than the traditional model would suggest. We also have provided a reference implementation of our sensor model in Python, `optika.sensors`, to make this noise model simple to integrate with existing instrument data processing pipelines.

This work does not consider read noise or other noise sources since those will be specific to the particular camera electronics of each instrument, and also cannot be characterized using a VMR. Instead, this work focuses only on the noise intrinsic to the charge-generation process, since this is comparatively consistent across different instruments.

We plan to use this work to model the noise for the EUV Snapshot Imaging Spectrograph (ESIS) (J. D. Parker et al. 2022), as well as the Full-sun Ultraviolet Rocket Spectrometer (FURST), sounding-rocket-based spectrographs developed by our research group for observing the solar atmosphere.

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NASA grants XXXX, XXXX and XXXX. This article is an example of an executable paper (J. Lasser 2020), all of the simulations, figures, and tables are generated dynamically when this article is created. This helps prevent mistakes and allows this research to be as repeatable as possible. The code to create this document is available at <https://github.com/roytsmart/ccd-noise-paper>.

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